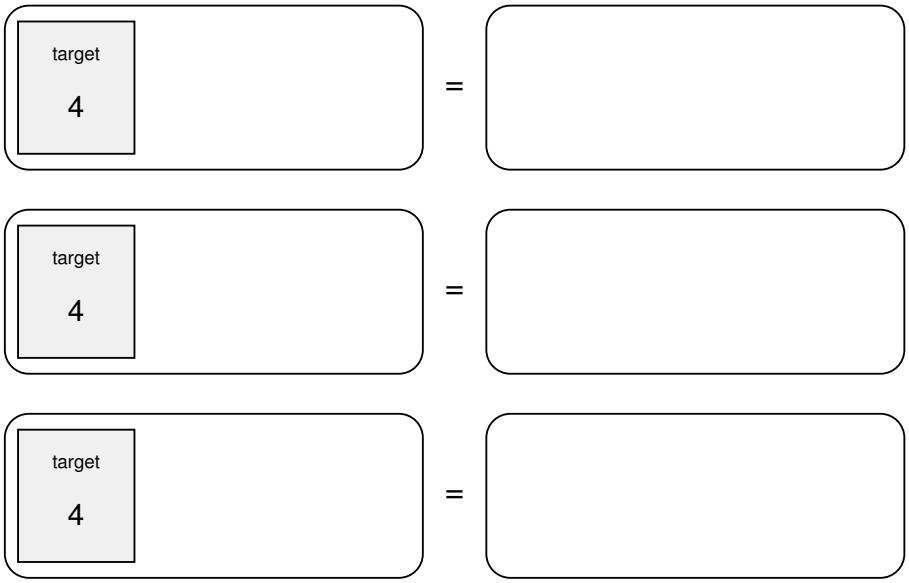
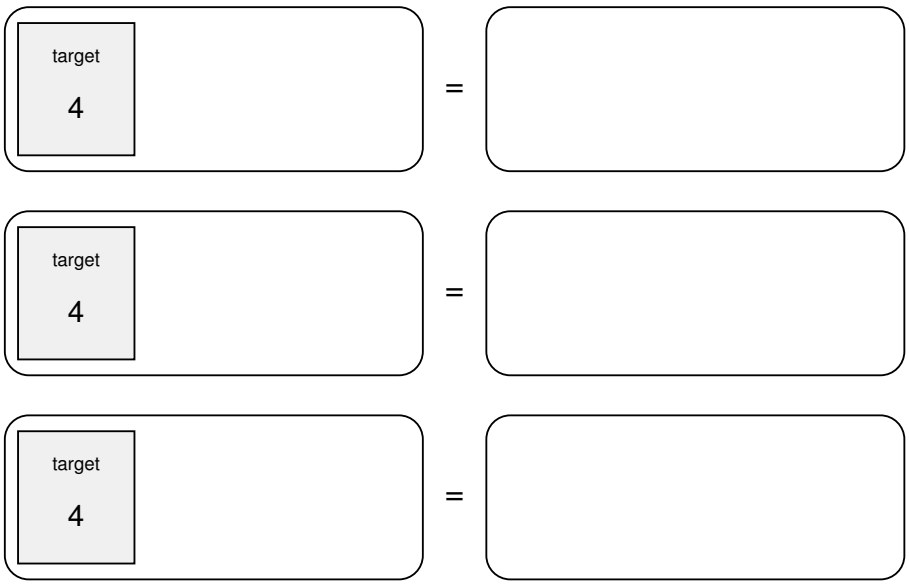
Problem 4: Use weights 1, 3 and 8 to balance 4 in every possible way. Then find every way to balance 4 with 1, 3 and 9.

1, 3, 8



1, 3, 9



Problem 5: Could a different three-weight kit balance every target from 1 through 14? Find one or explain why there cannot be one.

Problem 6: Choose a fourth weight to add to 1, 3 and 9. Make the longest possible unbroken run of targets starting at 1, and explain why your kit reaches all of them.